PSY221A: Final Data Validation and Wrangling

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#1 Hypotheses

H1: Individualistic attitudes (DV) will be stronger among full-time employed individuals than among unemployed individuals (IV).

H2: Gender-equality attitudes (DV) will be associated with daily time spent enhancing appearance (IV).

#2 Load the package and data

```
library(codebook)
library(here)

raw <- read.csv(here("data/raw/data_doctored.csv"))

# Keep only variables needed for H1 and H2
```

```r
data <- raw[, c(
  "ResponseId",
  "Gender_Equality_1", "Gender_Equality_2", "Gender_Equality_3",
  "Time_A_makeup_1", "Time_A_hyg_1", "Time_A_cosm_1",
  "Time_A_exercise_1", "Time_A_hair_1", "Time_A_dress_1",
  "Time_A_diet_1", "Time_A_other_1",
  "Individualism_col_1", "Individualism_col_2",
  "Individualism_col_4", "Employment"
)]
```

3. Validate participant IDs

Check for missing, blank, and duplicated participant IDs. Remove rows without a usable participant ID.

```r
# Count missing IDs
sum(is.na(data$ResponseId))

# Find in which row ID is missing
which(is.na(data$ResponseId))

#Check if there are more  missing IDs
data$ResponseId[data$ResponseId %in% c(-99, 99, 0, "N/A", "", " ")]

# Count duplicated IDs
sum(duplicated(data$ResponseId))

# Output: 1 NA for ReponseId

# Remove row that is NA
data <- data[!is.na(data$ResponseId),]

# Check the dimensions after cleaning
dim(data)

#Output: One less particpant (the NA got removed)
```

Hypothesis 1: Individualism and employment

1. Validate and recode employment

- Check for missing values in the Employment Variable
- The Employment variable allows multiple selections. Valid values for H1 are those who selected either 1 (=Full Time) or 4 (= Unemployed), but not both (mutually exclusive).
- Important: This includes people who selected Full-time plus another option, such as Student. So, additional selections are allowed, just selecting both Full-time and Unemployed leads to exclusion

```r
# Inspect Employment variable
unique(data$Employment)
sum(is.na(data$Employment))
sum(data$Employment =="", na.rm=TRUE)
range(data$Employment,na.rm = TRUE)
str(data$Employment)


# Missing Values
## Set blanks to NA
data$Employment[trimws(data$Employment) == ""] <- NA

## Create logical condition 1 for NAs of the Employment variable
missing_employment <- is.na(data$Employment)


# Invalid Values
## Create logical condition 2 for invalid inputs of the Employment variable
## Invalid are those  who selected both unemployed and full time
both_selected <- grepl("(^|,)1(,|$)", data$Employment) &
  grepl("(^|,)4(,|$)", data$Employment)

# Create the full-time employment variable using if else ()
## the grepl part of the function turns the TRUE FALSE into a binary 1/0
data$Full_time <- ifelse(
  missing_employment | both_selected,
  NA,
  as.integer(grepl("(^|,)1(,|$)", data$Employment))
)

# Create the unemployment indicator
data$Unemployed <- ifelse(
  missing_employment | both_selected,
  NA,
  as.integer(grepl("(^|,)4(,|$)", data$Employment))
)

# Check how many participants selected both categories
sum(both_selected, na.rm = TRUE)

# Check the frequencies for each variable where 1 = fulltime/ unemployed
table(data$Full_time, useNA = "ifany")
table(data$Unemployed, useNA = "ifany")

# Check whether anyone is coded as both
table(data$Full_time, data$Unemployed, useNA = "ifany")

# Create a new variable (Employment_group) that classifies participants into full-time vs. unemployed
# Nested ifelse() function: If TRUE assigns full time, if FALSE move on to next ifelse; those that are
data$Employment_group <- ifelse(
  data$Full_time == 1,
  "Full-time",
  ifelse(data$Unemployed == 1, "Unemployed", NA)
```

```
)

# Set Employment Group to a factor
## Gives categories an order
## Levels argument specifies the order of those categories
data$Employment_group <- factor(
  data$Employment_group,
  levels = c("Unemployed", "Full-time")
)
```

2. Validate and score individualistic attitudes

Use Individualism items 1, 2, and 4. Convert responses to numeric values, treat special missing codes as missing, check the valid response range, and reverse-code the items. Higher composite scores represent greater individualism.

```
# Create an object for the columnnames
ind_var <- c(
  "Individualism_col_1",
  "Individualism_col_2",
  "Individualism_col_4"
)

# Inspect data and data structure
##Inspect like for Employment variable
## Using lapply, indexing the variable columns, function = head to get preview of data for all three
lapply(data[ind_var], head)
## Using sapply, indexing the variables, function = class to get the data type
sapply(data[ind_var], class)


# Convert responses to numeric and set missing values to NA
data[ind_var] <- lapply(data[ind_var], function(x) {
  x <- trimws(x)
  x[x %in% c("", "-99", "N/A")] <- NA
  as.numeric(x)
})


# Check for invalid responses outside 1-7 or not whole numbers
sapply(data[ind_var], function(x) {
  sum(!is.na(x) & (x < 1 | x > 7 ))
})

# Set invalid responses to NA
data[ind_var] <- lapply(data[ind_var], function(x) {
  invalid <- !is.na(x) &
    (x < 1 | x > 7)
  x[invalid] <- NA
  x
})

# Reverse-code the items by 8-x
```

```r
## Creates new columns for the reverse coded columns
data[paste0(ind_var, "_R")] <- 8 - data[ind_var]

# Composite variable: Calculate the mean of all three revere coded items; Missing items result in compo
## Here paste is used to select the same columns that I created before
data$ind_rev_mean <- rowMeans(
  data[paste0(ind_var, "_R")],
  na.rm = FALSE
)

# Inspect the composite scores
summary(data$ind_rev_mean)
```

3. Create the H1 analysis dataset

Create one employment-group variable so that only full-time employed and unemployed participants are included. Participants in other employment categories, with missing employment information, or with conflicting responses are excluded.

```r
# Create a smaller dataset for H1
## Remove NAs for Individualistic Attitude and Employment Variable (cleaned up variables from before) u

h1_data <- data[
  !is.na(data$ind_rev_mean) &
    !is.na(data$Employment_group),
  c("ResponseId", "Employment_group", "ind_rev_mean")
]

# Check the final H1 dataset
dim(h1_data)
head(h1_data)
table(h1_data$Employment_group, useNA = "ifany")
colSums(is.na(h1_data))
```

Hypothesis 2: Gender Equality and Appearance Time

1. Validate and score Gender Equality Attitudes

Convert the three gender-equality items to numeric values, handle missing codes, and check the valid response range. The items are reverse-coded so higher scores indicate stronger gender-equality endorsement, as confirmed against the codebook.

```r
# Create a vector for the Gender Equality items
ge_var <- c(
  "Gender_Equality_1",
  "Gender_Equality_2",
  "Gender_Equality_3"
)

# Inspect the the variable values
```

```r
lapply(data[ge_var], function(x) sort(unique(x)))
lapply(data[ge_var], function(x) sum(is.na(x)))
lapply(data[ge_var], function(x) sum((x)=="",na.rm=TRUE))
sapply(data[ge_var], function(x) range((x),na.rm=TRUE))
lapply(data[ge_var], function(x) str ((x)))
## Output: range & str show that values are stored as characters not numerics


# Convert responses to numeric and handle missing codes
data[ge_var] <- lapply(data[ge_var], function(x) {
  x <- trimws(x)
  x[x %in% c("", "-99", "N/A")] <- NA
  as.numeric(x)
})

#Check if succesful
lapply(data[ge_var], function(x) sort(unique(x)))

# Check for invalid responses outside 1-7 or not whole numbers
sapply(data[ge_var], function(x) {
  sum(!is.na(x) & (x < 1 | x > 7))
})
## Output = 0

# Reverse-code the items
data[paste0(ge_var, "_R")] <- 8 - data[ge_var]

# Calculate the mean; if one of the three items is NA, the composite variable becomes NA
data$Gender_Equality_mean <- rowMeans(
  data[paste0(ge_var, "_R")],
  na.rm = FALSE
)

# Inspect the composite scores
summary(data$Gender_Equality_mean)
```

2. Validate and score appearance-enhancement time

The eight variables measure minutes per day spent on appearance-related activities. A value of 360 represents six hours or more for an individual activity and is retained as valid. The total is calculated only when all eight activity responses are available.

```r
# Create a vector with the column names of Appearance Time items
time_var <- c(
  "Time_A_makeup_1",
  "Time_A_hyg_1",
  "Time_A_cosm_1",
  "Time_A_exercise_1",
  "Time_A_hair_1",
  "Time_A_dress_1",
  "Time_A_diet_1",
  "Time_A_other_1"
```

```
)

#Inspect the data
lapply(data[time_var], function(x) sort(unique(x)))
lapply(data[time_var], function(x) sum(is.na(x)))
lapply(data[time_var], function(x) sum((x)=="",na.rm=TRUE))
## output: no blanks
sapply(data[time_var], function(x) range((x),na.rm=TRUE))
##output: range for each is 0-360
lapply(data[time_var], function(x) str ((x)))
sapply(data[time_var], class)


# Count invalid values below 0 or above 360
sapply(data[time_var], function(x) {
  sum(!is.na(x) & (x < 0 | x > 360))
})
# Output: no invalid scores

# Final Appearance Time Variable: Calculate Sum of individual Items

## Using mean prorating to maintain those participants who have less than or max.two NAs
## Do this, so I don't loose too much data

#Create logical condition 1: those, not NA
n_time_answered <- rowSums(!is.na(data[time_var]))

#Calculate mean, for mean prorating
time_mean <- rowMeans(data[time_var], na.rm = TRUE)

#Condition it checks against is if NAs =< 2, if yes calculate estimated score if they had filled out al
data$appearance_time_sum <- ifelse(
  n_time_answered >= 6,
  time_mean * length(time_var),
  NA
)
#Check if total time spent across activities exceeds 24hrs
sum(data$appearance_time_sum > 1440, na.rm=TRUE)
#output: 1692 have total >1440

#Set those >1440 to NA
data$appearance_time_sum <- ifelse(
  data$appearance_time_sum > 1440,
  NA,
  data$appearance_time_sum
)

# Check it worked
sum(data$appearance_time_sum > 1440, na.rm = TRUE)

# Inspect the total scores
## Now max is 1440 (24hrs)
summary(data$appearance_time_sum)
```

3. Create the H2 analysis dataset

Keep only participants with Scores (not NA) for both gender-equality attitudes and total appearance-enhancement time.

```
h2_data <- data[
  !is.na(data$Gender_Equality_mean) &
    !is.na(data$appearance_time_sum),
  c(
    "ResponseId",
    "Gender_Equality_mean",
    "appearance_time_sum"
  )
]

# Check the final H2 dataset
dim(h2_data)
head(h2_data)
colSums(is.na(h2_data))
```

Add variable labels

The labels describe what each variable measures, how composite scores are calculated, and the relevant units. Keep the labels in the same order as the columns.

```
# Labels for H1
var_label(h1_data) <- c(
  "Unique participant ID (ResponseId)",
  "Employment Comparison Group (Employment_group): Full-time or Unemployed; other employment categories
  "Individualistic Attitudes (ind_rev_mean): Response range for each item 1-7; mean of Individualism_co
)

# Labels for H2
var_label(h2_data) <- c(
  "Unique participant ID (ResponseId)",
  "Gender Equality Attitudes(Gender_Equality_mean): mean of Gender_Equality_1, Gender_Equality_2, and G
  "Daily Time Spent Enhancing Appearance(appearance_time_sum): Sum of eight Time_A variables covering ma
)
```

Create codebooks

The descriptions come from the original codebook and our scoring decisions. `compact_codebook()` adds summaries from the actual data. Knit the Rmd to display the formatted codebooks.

```
#generate the h1_data codebook
compact_codebook(h1_data)

#generate the h2_data codebook
compact_codebook(h2_data)
```

Save the final tables

```
# Create the output folder if it does not already exist. If it doesnt: dir.create("data/processed_data"


# Save the cleaned H1 and H2 datasets
write.csv(
  h1_data,
  "data/processed_data/h1_revised.csv",
  row.names = FALSE
)

write.csv(
  h2_data,
  "data/processed_data/h2_revised.csv",
  row.names = FALSE
)

# Read the files back in to verify they were saved
h1Saved <- read.csv("data/processed_data/h1_revised.csv")
h2Saved <- read.csv("data/processed_data/h2_revised.csv")

# Final Check dimensions and missing values
dim(h1Saved)
dim(h2Saved)

colSums(is.na(h1Saved))
colSums(is.na(h2Saved))
```